

Financial Fragility, Structural Instability, and Portfolio Investment: Evolutionary Dynamics Revisited*

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Abstract

This study examines the evolutionary aspects of portfolio investment involving risky and risk-free assets in an uncertain market environment. We consider an investment game in which profit maximization by adjusting the number of risky assets in the portfolio is the basis of investment dynamics, similarly found in the governing rule for resource competition between hawks and doves in population dynamics. The investment dynamics exhibit a stable Nash equilibrium in which investors who hold as many risky assets as feasible accumulate more wealth in the repeated game. Since an increase in individual investor wealth owes much to risk-taking behavior, investment dynamics are susceptible to recurrent bubbles and crises. Given that wealth accumulation directly results from increased risky assets, the portfolio choice for risky assets seems rational (i.e., optimal) when investors face uncertainty. This situation provides insight into why financial institutions and investors are exposed to financial fragility and structural instability during periods of economic uncertainty.

JEL Classification: C63; C73; E32; G11

Keywords: Bubbles; Financial fragility; Hawk–Dove game; Portfolio choice; Structural instability

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Significance

This study considers the hawk-dove game to capture the significance of risk-return trade-off in portfolio choice. Preserving simplicity in investment decisions based on an evolutionary game, we model a governing guideline for the decision rule that investors tend to obey. The investment dynamics with optimal asset allocation will converge to a stable Nash equilibrium, in which investors who hold as many risky assets as feasible accumulate more wealth in the repeated game. We show that competitive markets drive the investment dynamics to the boundary between a normal state and a crisis state. Consequently, collective decisions of rational agents increase their financial fragility and have a negative effect on systemic risk, making the economy inherently vulnerable to crisis.

1 Introduction

Financial markets exhibit persistent episodes of boom and collapse that standard rational models struggle to explain. While speculative behavior, leverage, and herding have been identified as drivers of crises [see, e.g., Brock and Hommes (1), Cont and Bouchaud (2), Day and Huang (3), De Long et al. (4), Gaunersdorfer et al. (5), Kirman (6), Lux (7)], less attention has been paid to the evolutionary mechanism underlying investors' strategic adaption [see also Amir (8)]. We propose a parsimonious evolutionary model in which wealth accumulation itself acts as the fitness criterion for survival in financial markets. Investors who allocate more to risky assets earn higher returns during stable periods and thus gain greater market influence. However, exceeding the critical threshold is associated with elevated systemic vulnerability and an increased probability of large drawdowns – an evolutionary manifestation of Minsky's notion that *stability breeds instability* [see Minsky (9)].

Our approach adopts the hawk-dove framework [see, e.g., Chen et al. (10), Hanauske et al. (2010)] and tailors it to portfolio choice with the risk-return trade-off: high but unstable return for risky investment (hawk strategy) versus low but stable return for risk-free investment (dove strategy). Profit maximization by adjusting the number of risky assets in the portfolio is the basis of investment dynamics, as is similarly found in the governing rule for resource competition between hawks and doves in population dynamics. Note here that investors with a successful

strategy attract more peers, while reducing prospective returns from a less successful strategy may discourage investment. Consequently, the replication process in the agent population, x_i , can be modeled using the differential equation:

$$\dot{x}_i = x_i(\pi_i - \bar{\pi}),$$

where $\dot{x}$ is the time derivative of population. π_i and $\bar{\pi}$ are the payoff of strategy i and the average payoff, respectively. This mechanism lays a foundation for individual portfolio choice with macro-level structural instability, showing how competition can lead to instability of the economic system [see also Vercelli (12)].

In particular, we consider the investment strategy as a partition of endowment in a continuum space of portfolio choice with possible actions to make the investment dynamics consistent with the description in financial markets. Moreover, we treat the returns from the previous game as an endowment for investment in the next period (cumulative payoff). As a novel feature of the modifications in the investment game, we verify that the optimal portfolio choice is consistent with an evolutionary stable Nash equilibrium.

In this game, the crucial factor that each agent must consider is the fact that the macroeconomic state whether the aggregate risk ratio exceeds a tipping point. If the ratio does not exceed the point regardless of the agent's portfolio, the best strategy of the agent is to invest all endowment to the risky asset. For the same reason, if the risk ratio exceeds the tipping point for all feasible portfolios of the agent, the best strategy is to hold a safe asset [see Cho and Jang (13)]. Hence, investors who hold as many risky assets as feasible accumulate more wealth in the repeated game. In other words, the higher rate of return from the risky asset contributes to an increase of the total wealth over time, and Nash equilibrium emerges in the long run, where agents should hold as many of the risky assets without exceeding the threshold where crisis occurs.

The structural instability and high risk in the model can be compared to the bubbles of Allen and Gale (14). In their model, overinvestment in the risky asset drives the asset price away from the fundamental value and creates bubbles. In our model, the risk-return trade off emphasizes the high volatility in the risky asset, and the risky strategy undermines the financial stability in the economy causing the bubble to burst. Indeed, the approach in this study shares similarities

with the literature, such as behavioral traits acting as catalysts for economic fluctuations. However, we pay more attention to the evolutionary nature of the decision-making process for investment [see, e.g., Maynard Smith (15), Hofbauer and Sigmund (16)]. As a merit, our game-theoretic approach to investment does not require strict assumptions, apart from information about return rates available to investors. The focus is on the effect of portfolio choice on wealth dynamics, where recurrent bubbles and disruptions are partly attributed to risk-taking bias. This approach enables us to approximate the dynamic behavior of many investors, and the investment dynamics can be seen as a natural approximation to bubbles based on optimal asset allocation under uncertainty.

The focus on Nash equilibrium is crucial in this context. In standard asset-allocation models, equilibrium is typically associated with stability and optimal risk sharing. In contrast, our framework demonstrates that when investment choices interact through aggregate risk exposure, Nash equilibrium can coincide with the maximum tolerable level of systemic risk. This result highlights a tension between individual optimality and collective stability, echoing classical insights from financial instability and game theory, but within a fully dynamic and evolutionary setting.

2 Hawk–Dove Game and the Financial Market

The main aspects of the Hawk-Dove Game (HDG) in financial markets take account of the volatility of investment dynamics. Initially, we consider Markowitz’s argument on risk-averse investors: investors face the risk–return trade-off in financial markets. A high-risk strategy has a higher but more unstable payoff, whereas a low-risk strategy leads to a lower but more stable payoff. In the simplest HDG (i.e., two players with pure strategy), we define Π_{AB} as the payoff when an agent’s strategy is A and the opponent’s strategy is B .

Table 1 shows the payoff matrices of the two agents in the HDG (left) and investment game (right). In HDG, when the strategy is H , an agent faces higher volatility in the payoff than with strategy D . In other words, the hawk strategy yields the highest payoff (Π_{HD}) when the opponent strategy is D , while it yields the lowest payoff (Π_{HH}) when the opponent strategy is H .

Conversely, the dove strategy has two payoffs: Π_{DD} and Π_{DH} . Its payoffs are lower than

Π_{HD} but higher than Π_{HH} . Overall, the 2-person HDG with pure strategy has a straightforward comparison of the payoffs: $\Pi_{HD} > \Pi_{DD} > \Pi_{DH} > \Pi_{HH}$. For simplicity, we assume that the payoff of strategy D is not affected by the opponent's decision: $\Pi_{DD} = \Pi_{DH}$.¹

Table 1: Payoff in 2-P Game: HD (Left) and Investment (Right)

	H_1	D_1		Risky ₁	Risk-free ₁
H_2	(Π_{HH}, Π_{HH})	(Π_{DH}, Π_{HD})	Risky ₂	$(-W_1, -W_2)$	$(r_S W_1, r_R W_2)$
D_2	(Π_{HD}, Π_{DH})	(Π_{DD}, Π_{DD})	Risk-free ₂	$(r_R W_1, r_S W_2)$	$(r_S W_1, r_S W_2)$

To set the stage for the investment game, we consider a direct mapping of HDG to portfolio choice. For example, R refers to the hawk strategy, which can be regarded as an investment in risky assets. The notation for S indicates the dove strategy, mapped into the investment in risk-free assets. Investors are assumed to enjoy a higher return from investment in risky assets ($r_R > r_S$); however, the conflict of risky strategies could result in a total loss of investments, i.e. $(-W_1, -W_2)$, where W stands for wealth. Note here that investors can be seen as opponents bidding for the risky assets, and thus driving the price up. The risk-taking bias in asset management could lead to highly leveraged strategies in portfolio choice, and the economy facing high market volatility and systemic risk is subject to bubble collapse.

Table 2: Main modifications of HDG

	Modification	Contents
(P1)	N -Person Game	Simultaneous game with massive investors
(P2)	Portfolio Strategy	Strategy profile as a ratio of risky and risk-free assets
(P3)	Cumulative Endowment	Wealth accumulation from previous periods
(P4)	Relative Payoff	Payoff as a form of interest rate from investment

¹This assumption stresses that the return on risk-free investment is hardly affected by economic fluctuations. Note that introducing this assumption does not change the main properties of HDG.

Table 2 summarizes the main modifications in a conventional HDG, which aim to create a consistent and realistic description of investment behavior in financial markets. First, the investment game assumes that many investors participate simultaneously (P1), which differs from the conventional evolutionary game model setting, where two agents are randomly selected from a pool of finite agents to play the game. This modification is because the financial market can be considered the comprehensive result of individual transactions of self-interested agents (or an arbitrarily large number of decentralized agents).

Second, the strategy profile of investors is represented by the ratio of risky assets in the portfolio choice (P2). Investors will not likely stick to pure or mixed financial market strategies. In pure strategies, investors choose between risky and risk-free assets but not both ("All-In" strategy). In reality, investors rarely behave in this manner. In a mixed strategy, investors randomly choose one of the pure strategies, which also does not match actual financial market behavior. Therefore, it is assumed in the game that investors can combine risky and risk-free assets in their portfolios.

Although our setting may appear superficially related to mixed strategy formulations, the interpretation and mechanics are fundamentally different. In mixed strategies, agents randomize over discrete actions, and equilibrium concepts rely on probabilistic choice. By contrast, we model investment behavior as a continuous portfolio decision, where each agent allocates a fixed share of wealth to the risky asset.

Importantly, individual portfolio shares are deterministic and time-invariant. The dynamics of aggregate investment arise solely from wealth accumulation and selection: strategies that yield higher realized returns grow in influence over time. This deterministic structure allows us to characterize equilibrium outcomes analytically and to isolate the role of aggregate risk exposure without introducing stochastic choice. Extensions that incorporate return volatility or behavioral noise are explored separately through numerical simulations.

Third, as an important means of evolution in financial markets (or "the principle of survival of the fittest"), the payoff of the previous game serves as the endowment in the next game (P3). When the game is repeated, the modification creates substantial wealth accumulation fluctuations, reflecting actual asset markets where endowments usually comprise payoffs that have grown through asset market transactions. In other words, asset prices are ultimately

formed due to an endowment-weighted purchase decision.

Fourth, we consider that the payoffs grow from the wealth multiplication by interest rates, and the accumulation of endowments allows for the divergence of wealth between investors in the repeated game (P4). Investors endowed with the same amount of money for their investment do not align with substantial fluctuations in wealth dynamics; this will be discussed later.

3 Model

Finite-agent formulation We first consider an economy with N investors indexed by $i = 1, \dots, N$. Each investor allocates a fixed share $\beta_i \in [0, 1]$ of her wealth $W_{i,t}$ to risky asset, and the remainder $1 - \beta_i$ to the risk-free asset. Let $r_R > r_S > 0$ denote the respective returns on risky and risk-free assets. The wealth-weighted average share of risky asset is

$$\hat{\beta}_t := \frac{\sum_i \beta_i W_{i,t}}{\sum_i W_{i,t}}. \quad (1)$$

When $\hat{\beta}_t \leq \bar{R}$, the economy is in a *normal state* and the risky asset yields r_R . When $\hat{\beta}_t > \bar{R}$, the market enters a *fragile state* in which risky asset returns drop sharply (e.g., -1). The threshold $\bar{R}$ thus represents the maximum level of aggregate risk the system can tolerate, analogous to a leverage or liquidity constraint in real markets.

Each investor's wealth then evolves multiplicatively as

$$W_{i,t+1} = W_{i,t}[1 + (1 - \beta_i)r_S + \beta_i g(\hat{\beta}_t)], \quad (2)$$

where $g(\hat{\beta}_t) = r_R$ in the normal state and $g(\hat{\beta}_t) = r_C = -1$ in the fragile state.

To cover the full range of strategies, we assign uniformly spaced initial values $\beta_i = (i - 1)/(N - 1)$. This grid ensures a balanced representation of risk preferences from fully safe to fully risky.

Continuum limit As $N \rightarrow \infty$, the discrete index i becomes a continuous label $x \in [0, 1]$, and the individual strategy β_i converges to a continuous function $k(x)$. Wealth $W_{i,t}$ is then written as $W_i(x)$, and the aggregate risky share becomes

$$\hat{\beta}_t = \frac{\int_0^1 k(x)W_t(x)dx}{\int_0^1 W_i(x)dx}. \quad (3)$$

The corresponding wealth dynamics follow

$$W_{t+1}(x) = W_i(x)[1 + (1 - k(x))r_S + k(x)g(\hat{\beta}_t)]. \quad (4)$$

A uniform initial distribution can be implemented by $k(x) = x$; more generally, one may specify a density $f(k)$ over $k \in [0, 1]$. This continuum representation simplifies the analysis of equilibrium and convergence while remaining consistent with the finite-agent interpretation.

Economic interpretation The wealth accumulation mechanism acts as the evolutionary selection rule: successful (i.e., high-return) strategies gain weight in aggregate wealth, tilting the system toward riskier exposure. This structure embeds the classical hawk-dove trade-off between high-risk/high-return and low-risk/low-return strategies. The threshold $\bar{R}$ serves as a coordination boundary where the aggregate equilibrium lies on the edge of fragility.

A central element of our framework is the existence of an upper bound $\bar{R}$ on the aggregate exposure to risky assets. This threshold should not be interpreted as an arbitrary modeling device. Rather, it captures a reduced-form representation of systemic constraints that emerge when collective risk-taking becomes excessive.

In modern financial systems, overinvestment in risky or speculative assets is often accompanied by leverage, margin requirements, and liquidity dependence. When aggregate exposure exceeds certain critical levels, even small shocks can trigger forced deleveraging, fire sales, and abrupt regime shifts in asset values. Such mechanisms have been extensively documented in the literature on financial instability, leverage cycles, and endogenous systemic risk.

In our model, $\bar{R}$ summarizes these institutional and market-based limits – such as risk-management constraints, liquidity shortages, or regulatory thresholds – beyond which the risky asset can no longer sustain its normal return-generating regime. Once the aggregate risky share crosses $\bar{R}$, the system transitions endogenously into a crisis state, in which the value of the risky asset collapses. This parsimonious formulation allows us to focus on how individually rational investment behavior can collectively push the system toward its own stability boundary.

4 Results

The model yields three main analytical results concerning equilibrium, convergence, and cyclical instability.

Proposition 1 (Nash Equilibrium in the One-Shot Game). *In a single-period game with N investors, β_i^* is the investor i 's best response if and only if*

$$\sum_i \beta_i^* \frac{W_i}{\widetilde{W}} = \bar{R},$$

where $\widetilde{W} = \sum_{i=1}^N W_i$ denotes *total aggregate wealth*. The term $W_i/\widetilde{W}$ thus represents investor i 's *relative wealth weight* in the economy. In equilibrium, each investor's best response collectively maintains the aggregate risky share at the critical threshold $\bar{R}$. This normalization ensures that the weighted sum of individual risky shares equals the aggregate exposure $\hat{\beta}_t$; hence, the equilibrium condition $\sum_i \beta_i^*(W_i/\widetilde{W}) = \bar{R}$ is equivalent to $\hat{\beta}^* = \bar{R}$.

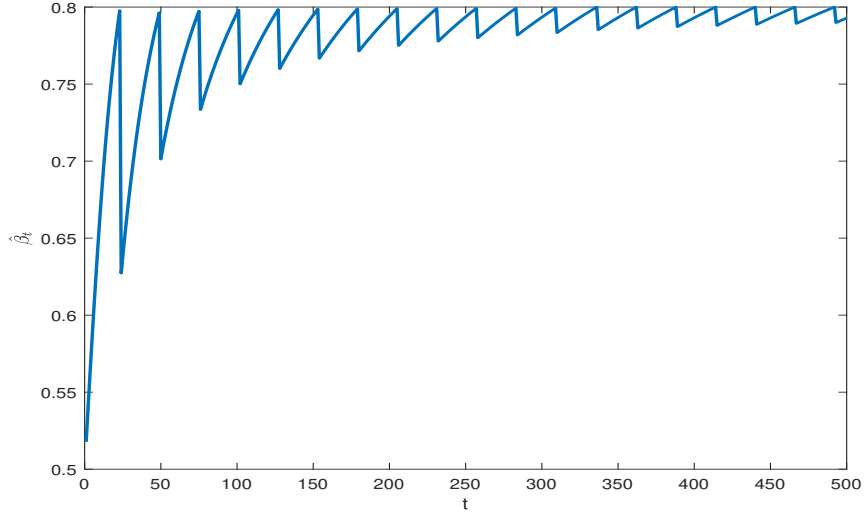
Economic interpretation The Nash equilibrium lies on the boundary separating the normal and fragile states. Investors implicitly coordinate at the edge of systemic stability: each finds it optimal to take on as much risk as possible, provided that aggregate exposure does not cross the threshold.

Proposition 2 (Dynamic Convergence in a Repeated Game). *In the repeated game with cumulative wealth, the wealth-weighted risk ratio $\hat{\beta}_t$ converges to $\bar{R}$:*

$$\lim_{t \rightarrow \infty} \hat{\beta}_t = \bar{R}. \tag{5}$$

Investors whose portfolios are closer to the optimal share accumulate wealth at higher rates and their relative influence increases. As a result, the aggregate risk share stabilizes endogenously at the fragility boundary.

Figure 1: Numerical simulation of $\hat{\beta}_t$ with $\bar{R} = 0.8$, $r_R = 0.1$, $r_S = 0.04$



Economic interpretation Wealth accumulation functions as an evolutionary selection rule. Successful strategies (ie, high-return) gain more weight in aggregate wealth, driving the system toward the maximum sustainable risk. This mechanism connects micro-level portfolio adaptation with macro-level financial fragility.

Proposition 3 (Crisis periodicity). *Let $\gamma_t = t/\Gamma_t$ denote the ratio of total periods to crisis occurrences up to time t . Then,*

$$\lim_{t \rightarrow \infty} \gamma_t = \frac{1 + r_R}{(r_R - r_S)(1 - \bar{R})}. \quad (6)$$

The average length of the normal phase increases with the safe return r_S and the threshold $\bar{R}$, but decreases with higher risky returns r_R . In other words, higher potential gains induce more frequent crisis.

Economic interpretation The system generates endogenous boom-bust cycles without external shocks. During stable phases, rising returns encourage greater risk-taking, pushing the aggregate share toward the threshold. When the limit is exceeded, the crisis phase resets wealth distributions, and the cycle repeats – an evolutionary manifestation of Minsky’s notion that sta-

bility breeds instability.

A risk-return trade-off does not, by itself, logically imply the existence of crisis states. Many standard models accommodate higher expected returns through increased volatility without regime changes. However, in financial markets, risk is often associated with nonlinear losses arising from leverage, liquidity constraints, and forced adjustments. In such environments, higher risk exposure can translate into discrete shifts in market conditions rather than smooth fluctuations.

Our model captures this empirically relevant feature by representing risk through a regime-switching structure. The crisis state should therefore be interpreted not as an inevitable consequence of risk-taking per se, but as a stylized representation of the systemic externalities that arise when aggregate exposure exceeds sustainable limits.

5 Numerical Simulation

This section considers numerical simulations for an investment game with 10,000 investors. Table 3 shows the parameter values in simulations and the game is repeated over 1,000 times.

Model A We consider some variations in the investment propensity (β_i) and initial wealth (ω_0) to check the robustness of the analytic results in the previous section.

Model B To assess the robustness of our results, we extend the baseline model by allowing the systemic threshold $\bar{R}$ to vary over time. The time-varying threshold $\bar{R}_t$ can be interpreted as reflecting changes in market liquidity, regulatory constraints, or aggregate risk tolerance. While $\bar{R}_t$ is treated as exogenous in the simulations, its evolution captures realistic fluctuations in the financial environment.

Model C We also relax the extreme assumption that the crisis-state return equals -1 . Allowing the crisis return r_C to lie strictly between -1 and the normal-state return r_S preserves the qualitative dynamics of the model. Even when the crisis state involves partial losses rather than complete collapse, the economy continues to evolve toward the systemic threshold, confirming that our results do not hinge on extreme payoff assumptions.

Table 3: Parameters in numerical simulation models

Parameter	Model A	Model B	Model C
Number of Investors	10,000	10,000	10,000
Crisis State Threshold ($\bar{R}$)	0.8	Varies	0.8
Return on Risky Asset (r_R)	$\{0.1, -1\}$	$\{0.1, -1\}$	$\{0.1, r_C\}, -1 \leq r_C < r_S$
Return on Risk-free Asset (r_S)	0.04	0.04	0.04

5.1 Model A

Table 4 summarizes the model settings for the numerical simulations. First, we consider the variations of initial wealth: the difference between Models 1 and 2 and between Models 3 and 4. Models 1 and 3 are equivalent to the numerical simulation of the theoretical model with multiplicatively cumulative wealth. Variations in investor portfolio choices are considered between Models 1 and 2 and between Models 3 and 4.

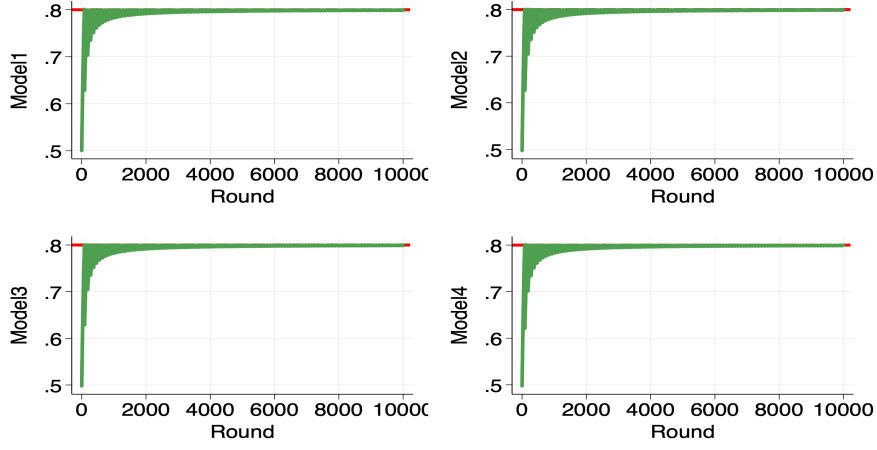
Table 4: Model setting in numerical simulations

Initial Wealth \ Risk Aversion	Equally Distributed	Random
Constant	Model 1	Model 2
Random	Model 3	Model 4

The Appendix collects the simulated distributions of investment propensity and initial wealth for the four models (see Figures V and VI). Figure 2 shows the dynamic behavior of average β_i in the repeated game. The simulation results verify that individual wealth and portfolio choice differences at the initial stage create little variation in the aggregated risky asset ratio. The aggregated risky asset ratio converges to the threshold value in all cases.

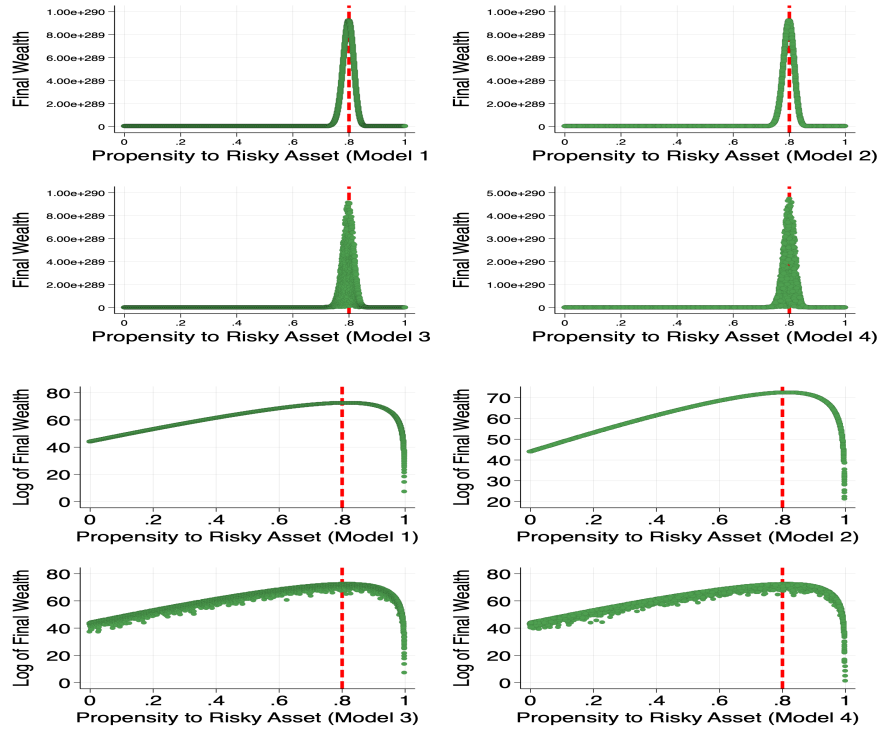
Figure 3 shows individual final wealth ($W_{i,t}$) scatter diagrams and the propensity to invest in risky assets (β_i). Note that T is the number of total rounds, and the final wealth is $W_{i,t}$, which verifies the results in Proposition 2: the optimal β_i converges to the threshold for the crisis state ($\bar{R}$), and relative wealth is concentrated on the investor with the optimal portfolio

Figure 2: Wealth-weighted average of β_i



choice.

Figure 3: Scatter diagram of final round



Note: Top panel for $W_{i,t}$ and β_i , Bottom panel for $\log(W_{i,t})$ and β_i

5.2 Model B

[Show the result of Simulation Model B]

5.3 Model C

[Show the result of Simulation Model C]

6 Discussion

This study examined the dynamic behavior of investment in a modified version of the hawk-dove game. We analyzed wealth dynamics from portfolio investment with many agents, and the game's simplicity allowed us to find a tractable solution for the optimal portfolio strategy. The portfolio choice of individual investors drives the system toward the boundary of financial fragility, and Nash equilibrium emerges with cycles of leverage buildup and collapse. Hence, the model provides a minimal evolutionary mechanism linking individual portfolio competition to macro-financial instability. $\bar{R}$. Even without irrational exuberance or external shocks, the economy endogenously generates .

Endogenous fragility Because investors' wealth represents their market influence, successful strategies amplify systemic exposure. This process transforms micro-level heterogeneity into macro-level fragility – an evolutionary form of Minsky's “stability breeds instability.”

Empirical relevance In empirical terms, the threshold $\bar{R}$ can correspond to observable indicators such as aggregate leverage ratios, risk-weighted assets, or market-wide Value-at-Risk limits. When these indicators tighten, our model predicts longer stable phases and smaller amplitude of cycles. Conversely, financial liberalization that relaxes such constraints shortens stability periods and increases systemic volatility.

Policy implications Macro prudential policies that cap leverage or increase safe-asset returns(r_S) can effectively lengthen stability cycles, but they may also slow wealth accumulation and widen cross-sectional inequality. Hence, financial stability and distributive equity may involve an inherent trade-off within this evolutionary setting.

Future extensions The current analysis assumes a fixed threshold $\bar{R}$. Allowing $\bar{R}$ to evolve endogenously with macro-financial conditions – such as changing liquidity or regulatory buffers

– will enable the study of how markets adapt to a moving fragility boundary. Preliminary simulations suggest that β_t tracks the shifting $\bar{R}_t$, maintaining the same boundary-following property found in the fixed-threshold case. A formal analysis of the extension is left for future work.

Overall, the evolutionary portfolio framework clarifies how self-reinforcing risk concentration, bounded by an adaptive threshold, can explain recurrent financial instability without exogenous shocks.

Appendix

A1: Proofs

[todo: rewrite by the changed notation and fix broken eq link] – after discussion 20251230

Proposition 1

Proof. The i^{th} investor seeks to maximize the payoff from his investment:

$$\max_{\beta_i} \Pi_i$$

where β_i^* is the best response in the investment game.

If all investors satisfy maximization ($i = 1, 2, \dots, N$), the strategy profile $\boldsymbol{\beta}^* := (\beta_1^*, \dots, \beta_N^*)$ is a Nash equilibrium. From Eqs. (??) and (1), we can rewrite Π_i into the function of β_i :

$$\Pi_i(\beta_i) = r_S W_i + \beta_i W_i \{ \mathbb{1}_{\{\hat{\beta} \leq \bar{R}\}} (2 + r_R) - (2 + r_S) \}.$$

The first-order condition is as follows:

$$\frac{\partial \Pi_i}{\partial \beta_i} = W_i \{ \mathbb{1}_{\{\hat{\beta} \leq \bar{R}\}} (2 + r_R) - (2 + r_S) \}. \quad (7)$$

The optimum value has three possibilities based on $\hat{\beta}$, and the cases under the normal or crisis states of the game have corner solutions:

- (1) **Normal state** ($\hat{\beta} \leq \bar{R}$) : We have $\frac{\partial \Delta W_i}{\partial \beta_i} > 0$ regardless of β_i ; hence $\beta_i^* = 1$. However, if $\hat{\beta} < \bar{R}$, the other investors have the possibility of $\exists \beta_j^{**} > \beta_j^*$ such that $\Pi_j(\beta_j^{**}) > \Pi_j(\beta_j^*)$. Therefore, the Nash equilibrium exists only when $\beta_i^* = 1$ and $\hat{\beta} = \bar{R}$.
- (2) **Crisis state** ($\hat{\beta} > \bar{R}$) : We have $\frac{\partial \Delta W_i}{\partial \beta_i} < 0$ regardless of β_i . However, if $\hat{\beta} > \bar{R}$, the other investors have the possibility of $\exists \beta_j^{**} < \beta_j^*$ such that $\Pi_j(\beta_j^{**}) > \Pi_j(\beta_j^*)$. Therefore, the Nash equilibrium exists only when $\beta_i^* = 0$ and $\hat{\beta} = \bar{R}$.
- (3) **Transition state** : The value of β_i decides the crisis. In other words, the risky investments of the investor determines the crisis state, and it makes $\bar{R} - \hat{\beta}$ either positive or negative. From Eq. (1), we consider the difference between the threshold and the aggregated risky

asset ratio:

$$\bar{R} - \hat{\beta} = \bar{R} - \frac{\beta_i W_i + \sum_{j \neq i} \beta_j W_j}{W_i + \sum_{j \neq i} W_j}.$$

Hence, $\bar{R} - \hat{\beta}$ is strictly decreasing in β_i because $\frac{\partial \bar{R} - \hat{\beta}}{\partial \beta_i} < 0$.

Suppose that $\beta_i^* = \frac{\bar{R} W_i + \sum_{j \neq i} (\bar{R} - \beta_j) W_j}{W_i}$, which makes $\bar{R} - \hat{\beta} = 0$. Then, it can be seen from Eq. (7) that $\Pi_i > 0$ if $\beta_i < \beta_i^*$ and $\Pi_i < 0$ if $\beta_i > \beta_i^*$. This outcome implies that $\beta_i = \beta_i^*$ maximizes W_i . Hence, $\hat{\beta} = \bar{R}$ is the Nash equilibrium.

□

Lemma 1 (Convergence Point of the Rate of Risky Asset Ratio).

In a cumulative PHDG, if $\hat{\beta}_t$ is convergent, it converges to $\bar{R}$.

Proof. From Eqs. (??), (??) and (??), we can get $\hat{\beta}_{t+1}$ in both states. For State 1,

$$\hat{\beta}_{t+1} = \frac{\int_0^1 k W_{i,t+1} dk}{\int_0^1 W_{i,t+1} dk} = \frac{\int_0^1 k [1 + r_S + k(r_R - r_S)] W_{i,t} dk}{\int_0^1 [1 + r_S + k(r_R - r_S)] W_{i,t} dk}. \quad (8)$$

Then we can express Eq. (8) in terms of $\hat{\beta}_t$:

$$\hat{\beta}_{t+1} = \frac{\tilde{r} \hat{\beta}_t + \mu_{2t}}{\tilde{r} + \hat{\beta}_t} = \frac{\tilde{r} \mu_{1t} + \mu_{2t}}{\tilde{r} + \mu_{1t}},$$

where $\tilde{r} = \frac{1 + r_S}{r_R - r_S}$, $\widetilde{W}_t = \int_0^1 W_{i,t} dk$, $\mu_{1t} = \hat{\beta}_t = \int_0^1 k \frac{W_{i,t}}{\widetilde{W}_t} dk$, $\mu_{2t} = \int_0^1 k^2 \frac{W_{i,t}}{\widetilde{W}_t} dk$.

In fact, $W_{i,t}(k)/\widetilde{W}_t$ is the density function of normalized wealth at time t , then $\mu_{1t} = \hat{\beta}_t$ can be interpreted as the first moment, and μ_{2t} can be interpreted as the second moment of wealth distribution at t . For State 2,

$$\hat{\beta}_{t+1} = \frac{\int_0^1 k W_{i,t+1} dk}{\int_0^1 W_{i,t+1} dk} = \frac{\int_0^1 k (1 + r_S)(1 - k) W_{i,t} dk}{\int_0^1 (1 + r_S)(1 - k) W_{i,t} dk}.$$

Then

$$\hat{\beta}_{t+1} = \frac{\hat{\beta}_t - \mu_{2t}}{1 - \hat{\beta}_t}.$$

From Cauchy-Bunyakovsky-Schwartz inequality for integrals,

$$\left[\int_0^1 \sqrt{W_{i,t}} k \sqrt{W_{i,t}} dk \right]^2 \leq \int_0^1 \left[\sqrt{W_{i,t}} \right]^2 dk \int_0^1 \left[k \sqrt{W_{i,t}} \right]^2 dk. \quad (9)$$

Dividing both side by $\widetilde{W}_t^2$, Eq. (9) can be expressed as follows:

$$\mu_{1t}^2 \leq \mu_{2t}.$$

The equality holds if and only if $\sqrt{W_{i,t}}$ and $k\sqrt{W_{i,t}}$ is linearly dependent. However, these two functions are not linearly dependent. Therefore, in State 1,

$$\hat{\beta}_{t+1} - \hat{\beta}_t = \frac{\mu_{2t} - \mu_{1t}^2}{S + \hat{\beta}_t} = \frac{Var(W_{i,t})}{S + \hat{\beta}_t} > 0. \quad (10)$$

And in State 2,

$$\hat{\beta}_{t+1} - \hat{\beta}_t = \frac{\mu_{1t}^2 - \mu_{2t}}{1 - \hat{\beta}_t} = -\frac{Var(W_{i,t})}{1 - \hat{\beta}_t} < 0. \quad (11)$$

Eqs.(10) and (11) implies $\hat{\beta}_t$ is oscillating around $\bar{R}$. Because $\hat{\beta}_t$ is bounded in $[0, 1]$, there exist at least a subsequence of $\hat{\beta}_t$ which converges to $\bar{R}$. From the Bolzano-Weierstrass theorem and the property of convergent sequence, any subsequence of the convergent sequence should converge to the same point and we can confirm that $\hat{\beta}$ converges to $\bar{R}$ if it is convergent. $\square$

Lemma 2 (Optimal Strategy of Fixed Cumulative PHDG). *Let Γ_t be a total number of crisis state up to period $t - 1$. Then the unique optimal portfolio strategy k_t^* of fixed cumulative PHDG exists:*

$$k_t^* = 1 - \frac{1 + r_R}{r_R - r_S} \frac{\Gamma_t}{t}.$$

Proof. Consider an investment game experiencing a crisis a total of Γ_t times over period $t - 1$. The wealth $W_{i,t}(k)$ at time t for strategy k can be expressed as a degree t polynomial in k , shown as:

$$W_{i,t}(k) = w_0[(1 + r_S)(1 - k)]^{\Gamma_t} [1 + r_S + k(r_R - r_S)]^{t - \Gamma_t}.$$

A unique critical point k_t^* is determined by the first-order condition:

$$\begin{aligned}\frac{dW_{i,t}(k)}{dk} &= w_0(1+r_S)^{\Gamma_t}(1-k)^{\Gamma_t-1}(1+r_S+k(r_R-r_S))^{t-\Gamma_t-1} \\ (t(1-k)(r_R-r_S) - \Gamma_t(1+r_R)) &= 0,\end{aligned}$$

leading to:

$$k_t^* = 1 - \frac{1+r_R}{r_R-r_S} \cdot \frac{\Gamma_t}{t}. \quad (12)$$

k_t^* also satisfies the second-order condition for maximum:

$$\left. \frac{d^2W_{i,t}}{dk^2} \right|_{k=k_t^*} = -(1-k_t^*)(1+r_S+k_t^*(r_R-r_S))\gamma_t(r_R-r_S) < 0.$$

□

Lemma 3 (Convergent Period in Fixed Cumulative PHDG). *The average period of investment game $\gamma_t = \frac{t}{\Gamma_t}$ is convergent.*

Proof of Lemma 3. It is equivalent to prove the sequence $\frac{\Gamma_t}{t}$ is convergent. To prove the convergence of $\frac{\Gamma_t}{t}$, it is sufficient to prove:

$$\forall \epsilon > 0, \quad \exists T \text{ s.t. } \forall t_2 > t_1 > T, \quad \left| \frac{\Gamma_{t_1}}{t_1} - \frac{\Gamma_{t_2}}{t_2} \right| < \epsilon.$$

From the given condition, we can confirm the below inequalities:

$$t_n > \Gamma_{t_n}, \quad n = 1, 2, \quad (13)$$

$$\Gamma_{t_2} \geq \Gamma_{t_1}, \quad (14)$$

$$t_2 - t_1 \geq \Gamma_{t_2} - \Gamma_{t_1}. \quad (15)$$

Suppose $T = \frac{t_2-t_1}{\epsilon}$, then for all $t_2 > t_1 > T$ and $\epsilon > 0$,

$$\frac{t_2 - t_1}{t_1} < \frac{t_2 - t_1}{T} = \epsilon.$$

- Case 1: $\frac{\Gamma_{t_1}}{t_1} \geq \frac{\Gamma_{t_2}}{t_2}$

$$\left| \frac{\Gamma_{t_1}}{t_1} - \frac{\Gamma_{t_2}}{t_2} \right| = \frac{t_2\Gamma_{t_1} - t_1\Gamma_{t_2}}{t_1t_2} \leq \frac{t_2\Gamma_{t_2} - t_1\Gamma_{t_2}}{t_1t_2} \quad (\text{From Eq. (14)})$$

$$= \frac{\Gamma_{t_2}(t_2 - t_1)}{t_1t_2} \leq \frac{(\Gamma_{t_1} + t_2 - t_1)(t_2 - t_1)}{t_1t_2} \quad (\text{From Eq. (15)})$$

$$< \frac{(t_1 + t_2 - t_1)(t_2 - t_1)}{t_1t_2} = \frac{t_2 - t_1}{t_1} < \epsilon. \quad (\text{From Eq. (13)})$$

- Case 2: $\frac{\Gamma_{t_1}}{t_1} < \frac{\Gamma_{t_2}}{t_2}$

$$\left| \frac{\Gamma_{t_1}}{t_1} - \frac{\Gamma_{t_2}}{t_2} \right| = \frac{t_1\Gamma_{t_2} - t_2\Gamma_{t_1}}{t_1t_2} < \frac{t_2\Gamma_{t_2} - t_2\Gamma_{t_1}}{t_1t_2} \quad (\because t_2 > t_1)$$

$$= \frac{\Gamma_{t_2} - \Gamma_{t_1}}{t_1} \leq \frac{t_2 - t_1}{t_1} < \epsilon. \quad (\text{From Eq. (15)})$$

□

Proposition 2

Proof. Let i_t^* be the agent index of k_t^* . From the lemma 2,

$$W_{i_t^*t}(k_t^*) > W_{i,t}(k) \quad \forall k \neq k_t^* \Rightarrow \frac{W_{i,t}(k)}{W_{i_t^*t}(k_t^*)} < 1 \quad \forall k \neq k_t^*.$$

From the Eq. (??),

$$\begin{aligned} \frac{W_{i,t}}{W_{i_t^*t}(k_t^*)} &= \frac{w_0 \{(1+r_S)(1-k)\}^{\Gamma_t} \{1+r_S+k(r_R-r_S)\}^{t-\Gamma_t}}{w_0 \{(1+r_S)(1-k_t^*)\}^{\Gamma_t} \{1+r_S+k_t^*(r_R-r_S)\}^{t-\Gamma_t}} \\ &= \left[\frac{1-k}{1-k_t^*} \right]^{\Gamma_t} \left[\frac{\tilde{r}+k}{\tilde{r}+k_t^*} \right]^{t-\Gamma_t} \\ &= \left[\left[\frac{1-k}{1-k_t^*} \right]^{\frac{\Gamma_t}{t}} \left[\frac{\tilde{r}+k}{\tilde{r}+k_t^*} \right]^{1-\frac{\Gamma_t}{t}} \right]^t < 1, \quad \forall k \neq k_t^*. \end{aligned} \quad (16)$$

where

$$\tilde{r} = \frac{1+r_S}{r_R-r_S}.$$

From the Lemma 3, $\frac{\Gamma_t}{t}$ is convergent and let $\frac{1}{\gamma}$ be the limit of $\frac{\Gamma_t}{t}$ and $\bar{k}$ be the limit of k_t^* . Then,

$$\lim_{t \rightarrow \infty} \frac{W_{i,t}}{W_{i,t}^*(k_t^*)} = \left[\left[\frac{1-k}{1-\bar{k}} \right]^{\frac{1}{\gamma}} \left[\frac{\tilde{r}+k}{\tilde{r}+\bar{k}} \right]^{1-\frac{1}{\gamma}} \right]^t.$$

From Eq.(16),

$$\left[\frac{1-k}{1-\bar{k}} \right]^{\frac{1}{\gamma}} \left[\frac{\tilde{r}+k}{\tilde{r}+\bar{k}} \right]^{1-\frac{1}{\gamma}} < 1, \quad \forall k \neq \bar{k}.$$

Therefore,

$$\lim_{t \rightarrow \infty} \frac{W_{i,t}}{W_{i,t}^*(k_t^*)} = \left[\left[\frac{1-k}{1-\bar{k}} \right]^{\frac{1}{\gamma}} \left[\frac{\tilde{r}+k}{\tilde{r}+\bar{k}} \right]^{1-\frac{1}{\gamma}} \right]^t = 0, \quad \forall k \neq k_t^*. \quad (17)$$

Eq. (17) implies the relative wealth of non-optimal strategy $k \neq k_t^*$ converges to 0 and also $Var(W_{i,t})$ converges to zero.

Therefore, for a sufficiently large t , $\hat{\beta}$ converges to a constant $\bar{k}$. Lemma 1 confirms that $\bar{k}$ should be equal to $\bar{R}$.

□

Proposition 3

Proof. From lemma 2, we can rewrite the Eq. (12) in terms of $\gamma_t = t/\Gamma_t$:

$$\gamma_t = \frac{1+r_R}{r_R+r_S} \frac{1}{1-k_t^*}.$$

Proposition 2 implies k_t^* converges to $\bar{R}$. Therefore:

$$\lim_{t \rightarrow \infty} \gamma_t = \frac{1+r_R}{r_R+r_S} \frac{1}{1-\bar{R}}.$$

□

A2: Distribution of β_i and initial wealth in the simulations

[todo: Model relabelling by compact]

Figure 4: Initial Distribution of Risky Asset Ratio (β_i) in Simulations

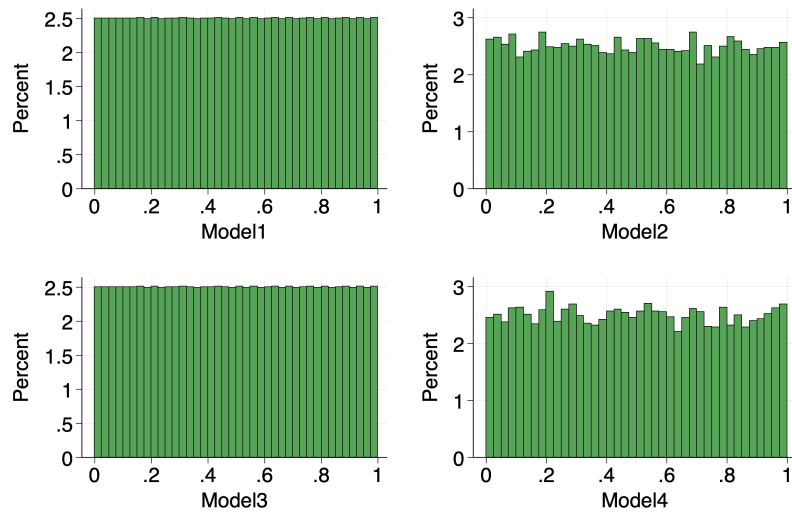
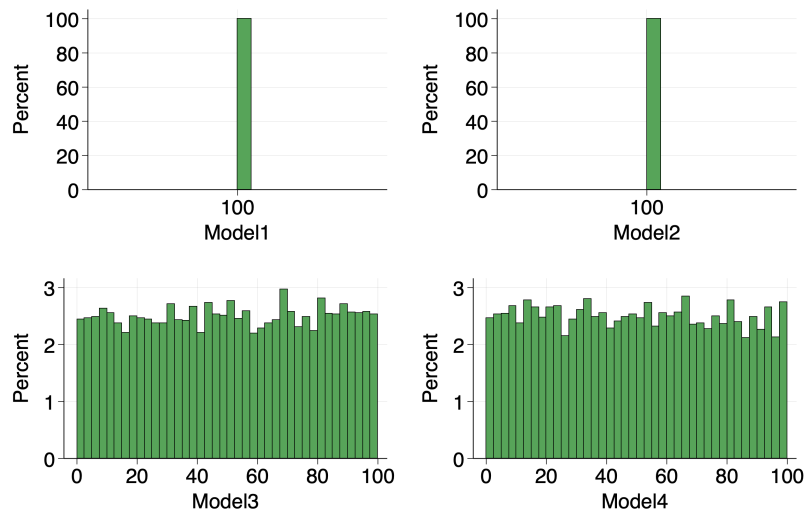


Figure 5: Initial Wealth Distribution in the Simulations



A3: Optimal portfolio (k_t^*), crisis period $\gamma_t = t/\Gamma_t$, and the ratio of maximum to total wealth in the simulations ($wealth^*/\widetilde{W}$)

Figure 6: k_t^* in the simulation

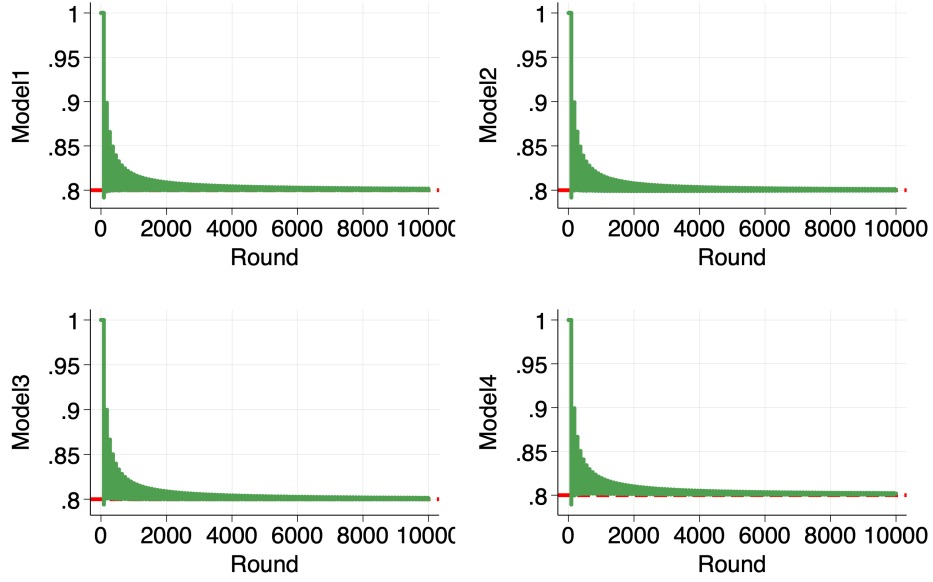


Figure 7: γ_t in the simulation

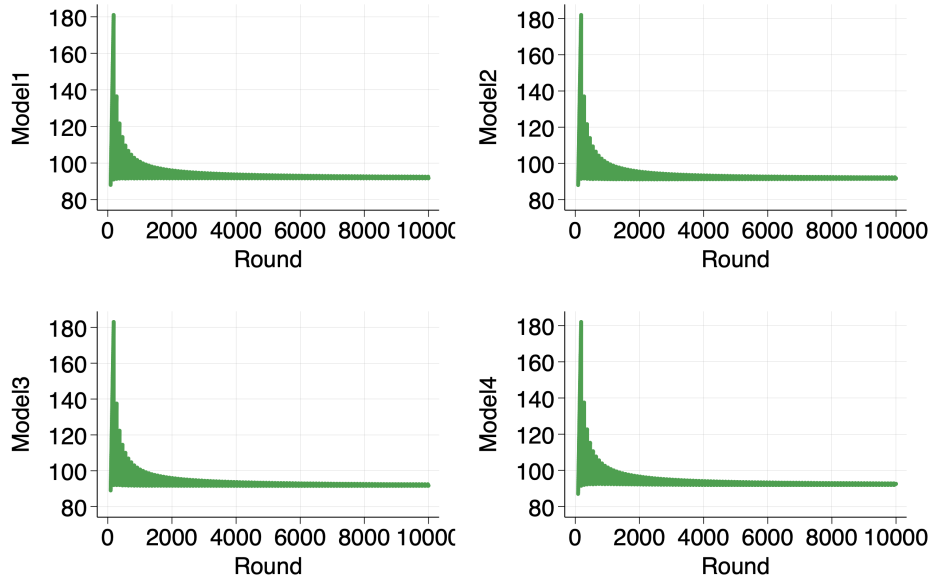
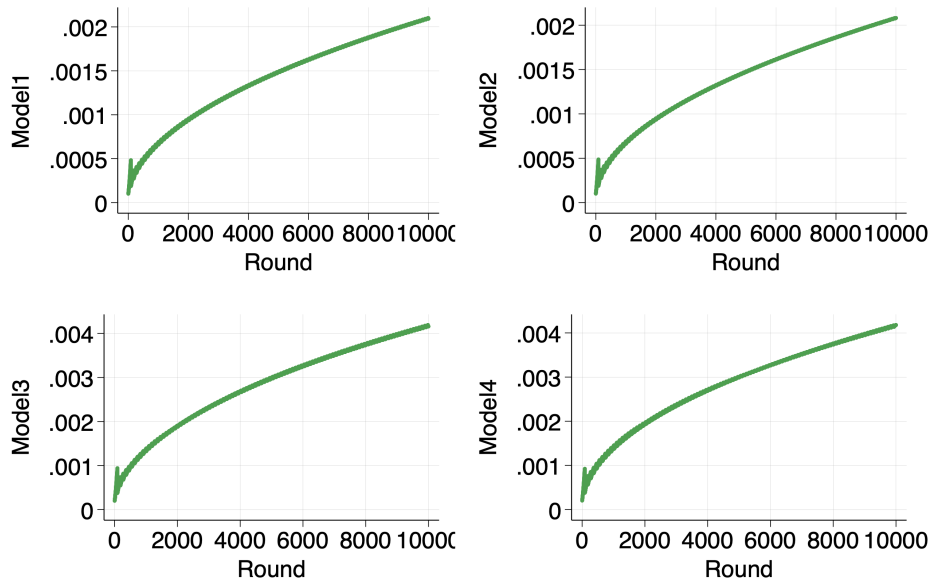


Figure 8: The ratio of maximum to total wealth in the simulation



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